Design proposal ilevie

Introduction

The goal of the design is to fabricate and measure two types of MZI circuits each one changed length difference and observe the change in free spectral range. The first MZI type uses y splitter at one side and directional couple on the other side, creating two outputs. This second MZI type uses y splitter at both it's input and output. Loop back circuit and y branch only circuit are also implemented for calibration. This work is a design proposal as a task for the EDx course UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis.

Theory

As explain in the course, the interferometer transfer function for both of the MZI described above is

$$\frac{i_{out}}{i_{in}} = \frac{1}{2}(1 + \cos(\beta \Delta L))$$

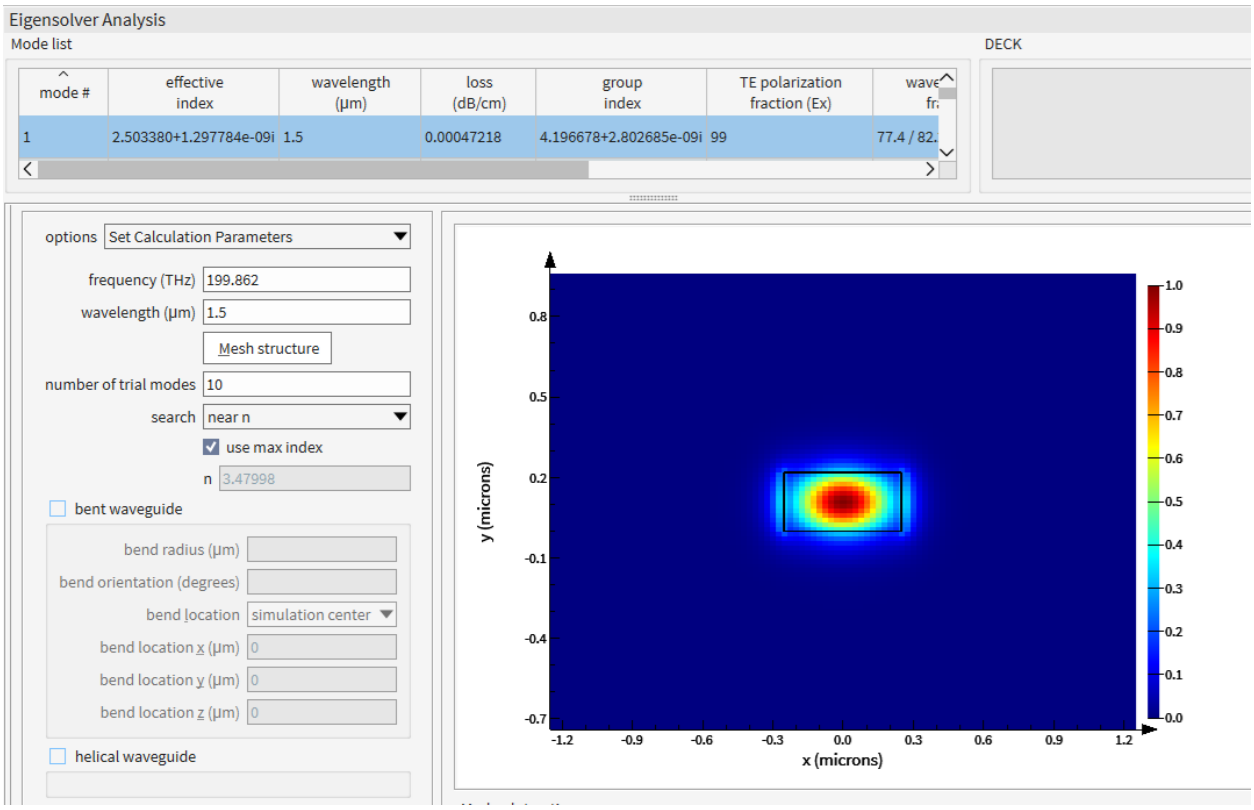
The free space spectral range (FSR) which is the spacing between adjacent peaks can be calculated to be

$$FSR = \frac{\lambda^2}{\Delta L n_g}$$

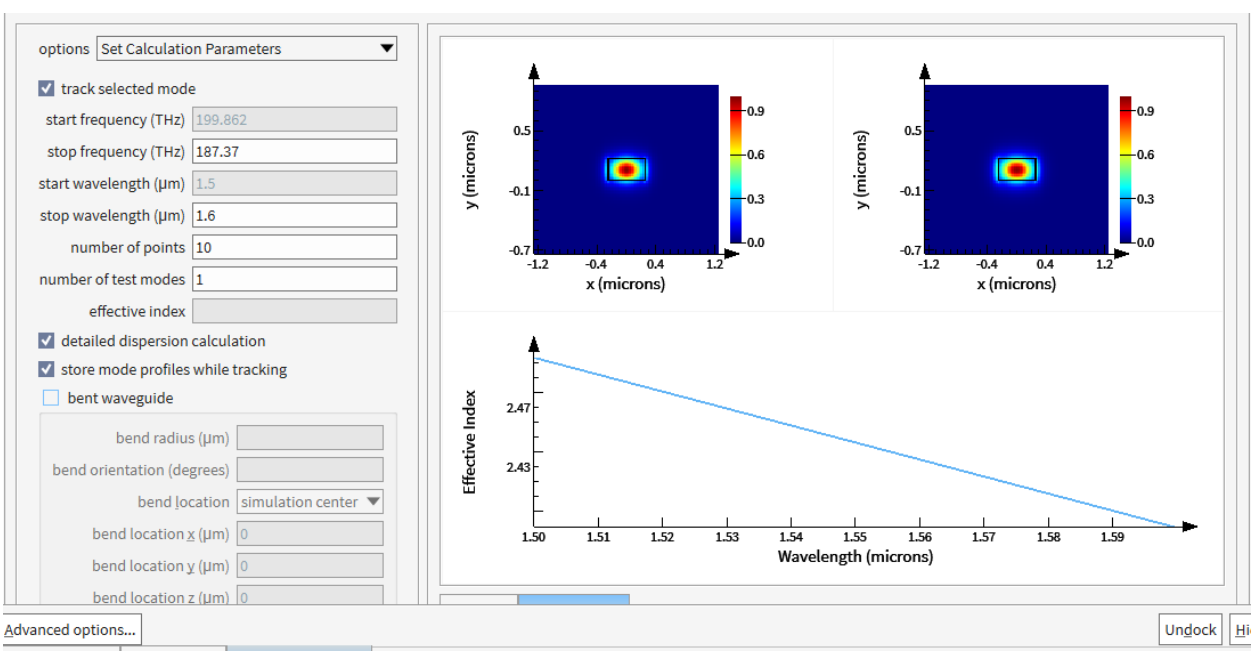
Thus by measuring the MZI FSR it is possible to extract the waveguide group index.

Selected waveguide geometry

The MZI is implemented using strip waveguide, TE polarized with width of 0.5um and height of 0.22um. The first TE mode profile and dispersion were calculated using Lumerical mode and are plotted next.



A frequency sweep was performed from 1.5um to 1.6um and the result was exported to MATLAB. A MATLAB script provided by the course was used to fit a compact model for the waveguide effective index



$$2.44682 - 1.1339 \cdot (\lambda - 1.55) - 0.0489866 \cdot (\lambda - 1.55)^2$$

Base on Taylor expansion

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

With relation to physical parameters as effective index, group index, and dispersion As described in the course notes.

$$n_{\text{eff}} = n_1$$

$$n_g = n_1 - n_2 \cdot \lambda_0$$

$$D = -2 \cdot \lambda_0 \cdot n_3 / c \quad [s/m^2]$$

Preliminary layout and design variation

There are total of seven MZIs implement.

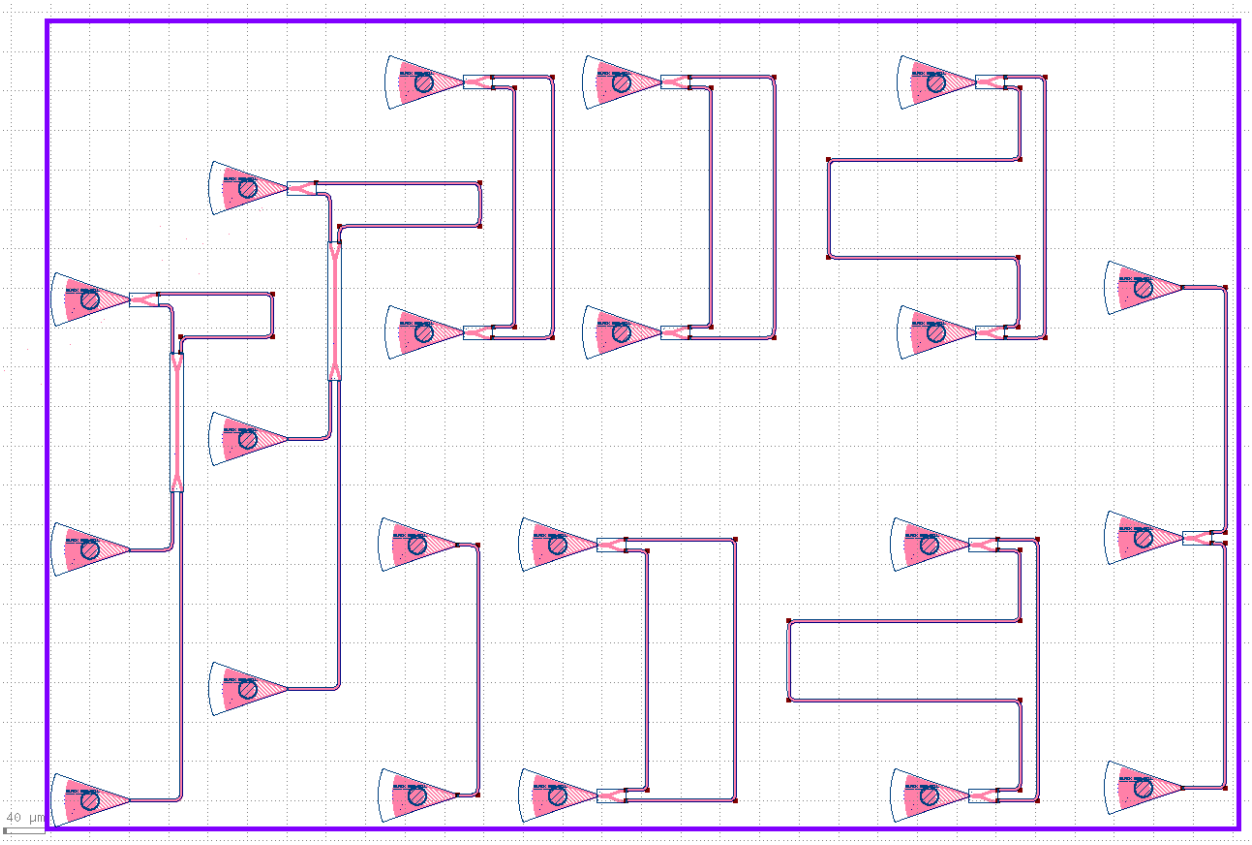
MZI1 and MZI2 are implemented using y splitter at the input and directional coupler at the output. The path difference is 100um and 150um.

MZI3 to MZI7 are implemented with y splitter at both input and output and path differences of 50um, 75um, 100um, 150um and 200um.

In addition a loop back is implemented to measure the grading coupler response and an circuit to measure the response of the y splitter used.

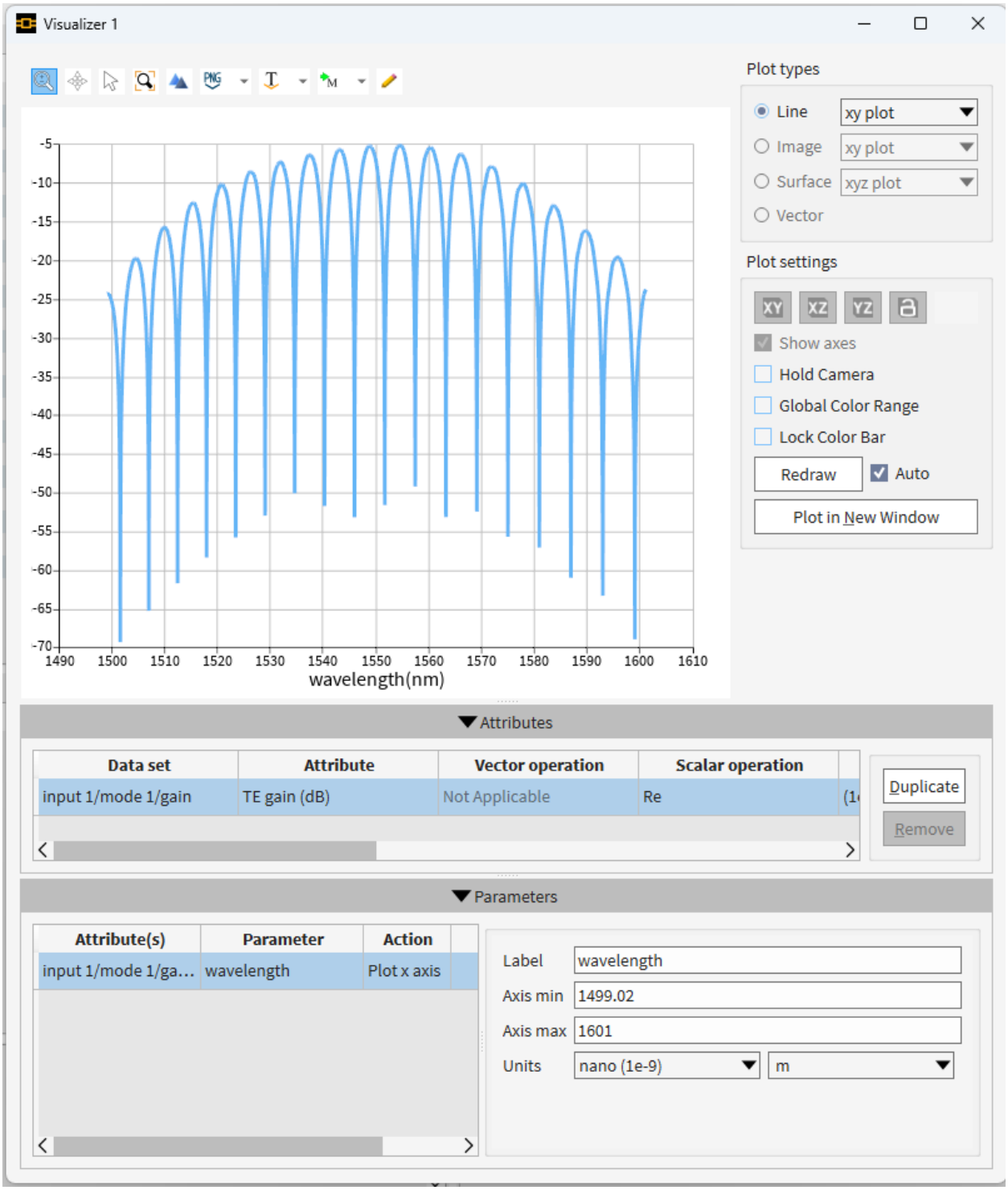
MZI number	path difference [um]	MZI Type
1	100	Y splitter plus coupler
2	150	Y splitter plus coupler
3	50	Y splitter only
4	75	Y splitter only
5	100	Y splitter only
6	150	Y splitter only
7	200	Y splitter only

Below is the layout as plotted in Klayout



Example transmission spectrum

Transmission spectrum of MZI5, which have path difference of 100um was simulated with Lumerical interconnect with below results



References

[1] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[2] L. Chrostowski, “Silicon Photonics Design, Fabrication and Data Analysis,” UBCx, edX. Accessed: May 4, 2026. [Online]. Available: https://www.edx.org/learn/engineering/university-of-british-columbia-silicon-photonics-design-fabrication-and-data-ana